though, let’s delve into the differences between these algorithms and A*.

## Listing 14.13 The A* algorithm

```
function aStar(graph, start, isGoal, distance,
heuristic)            #1
  queue ← new
PriorityQueue()                                          #2
  queue.insert(start,
0)                                                  #3
  distances[v] ← inf (" v ∈ graph | v <>
start)                       #4
  fScore[v] ← inf (" v ∈ graph | v <>
start)                           #5
  parents[v] ← null (" v ∈
graph)                                      #6
  distances [start] ←
0                                                   #7
  fScore[start] ← heuristic(start)
  while not queue.empty()
do                                                   #8
     v ←
queue.top()                                                    #9
     if isGoal(v)
then                                                      #10
       return v, parents
     else
       for e in graph.adjacencyList[v]
do                              #11
         u ← e.dest
         if distances[u] < distances[v] + e.weight
then               #12
           distances[u] ← distance(e) +
distances[v]                  #13
           fScore[u] ← distances[u] +
heuristic(u)                    #14
```